https://doi.org/10.31220/agriRxiv.2025.00295

Title: Accuracy in agricultural citizen science: farmers' assessment of harvest output in Ghana

Shortened title: *Accuracy in citizen science*

Name	Affiliations	Email	ORCID
Jacob Ulzen	1,2	julzen@ug.edu.gh	0000-0001-7378-1002
Nana Oduro Osei - Bonsu	2	gastone4gh@gmail.com	
George Bush	1,3	georgeadzibush@gmail.com	
Hugo Andrés Dorado Betancourt	4,5	hugo.dorado@wur.nl	0000-0002-0103-7505
Jacob van Etten	5*	j.vanetten@cgiar.org	0000-0001-7554-2558

Affiliations and addresses
1 CIAT, Accra, Ghana
2 Forest and Horticultural Crops Research Centre, College of Basic and Applied Science, School of Agriculture, University of Ghana, Nkwantanang, Ghana
3 Kookoo Pa Farmer Association, Kumasi, Ghana
4 Wageningen University, Wageningen, The Netherlands
5 Bioversity International, Montpellier, France
*Corresponding author

Accuracy in agricultural citizen science: farmers' assessment of harvest output in Ghana

Abstract

This study investigates methods for farmer-led yield estimation in decentralized on-farm trials in Ghana. Decentralized trials are key to reaching remote communities and reducing costs, but often require physical presence during time-sensitive tasks. Farmer participation in data collection, especially if data can be shared digitally, may help overcome these constraints. Farmers' accuracy in measuring harvest output is often unknown, and therefore leads to falling back on technicians. This study determines the accuracy of farmer measurement outputs and determinants of the accuracy by using simple on-farm exercises in which farmers execute the estimation and measurement tasks. We tested yield estimation methods, including volumetric measurement and weighing of maize kernels and cocoa beans, to evaluate farmers' accuracy. Volumetric measurement, involving a container of known volume, proved robust and accurate without prior training, performing consistently across farmers with various education levels. Weighing with scales, while theoretically more precise, was error-prone, even after training, due to issues with scales and taring. Refinements such as using electronic scales and obviating tare adjustments could improve weighing accuracy. Volumetric measurement emerged as a practical solution, especially where training is limited, offering consistent results. The results of this study provide important information to researchers on the type and the magnitude of error that they can expect when farmers measure harvest output. Our findings suggest that decentralized trials relying on farmer measurements can expand trial reach, reduce geographical bias, and optimize resource use by decreasing reliance on technician visits.

Abbreviations

RMSD: Root Mean Squared Deviation
MAD: Mean Absolute Deviation
MSD: Mean Signed Deviation
Tricot: Triadic comparisons of technology options

1. Introduction

Agricultural technologies such as new genotypes, agronomic techniques, next-generation inputs and other technologies need to be tested on farms to assess their performance, suitability, and acceptance by farmers under heterogeneous conditions. Farmers have long participated in on-farm

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

research, performing different roles. In some trial formats, farmers' roles are limited to provision of land, labor, and having general oversight of the field, and following instructions from researchers or technicians. In other trial formats, farmers have more responsibilities and participate in defining the goals, making agronomic decisions, and collecting and interpreting data (van Etten, 2011; Water-Bayers et al. 2015; Lacoste et al. 2022).

A limiting factor in decentralizing trials is the need for physical presence during time-critical tasks, especially near-simultaneous planting or harvesting of a large number of plots (van Etten, 2011; de Sousa et al. 2023). In addition, poor road networks make some communities inaccessible during parts of the year. To find locations for on-farm trials, researchers tend to select communities for convenience and accessibility, creating a bias against more remote communities (Ochieng et al., 2019). Active participation of individual farmers with oversight responsibilities in on-farm trials can remove or alleviate these constraints. If local data collection can be conducted by lead farmers and farmers can transmit data by mobile phone or through other digital media, travel costs can be reduced significantly.

Obtaining yield estimates is often a goal of on-farm trials. How to ensure that farmers produce accurate yield estimates from trial plots is still an open question. Yield estimates are necessary for variety release as evidence that candidate varieties yield as much as, or more than, the varieties they are expected to replace. Also, an important emerging goal of on-farm variety testing is to determine the performance of breeding programs. Progress estimates obtained from farms are now increasingly seen as the way to evaluate breeding operations focused on the Global South (Williamson and Leonelli 2022). This research was stimulated by the need to include metric yield estimates in tricot (triadic comparisons of technology options), a widely used approach. It involves farms as incomplete blocks (cf. Atlin et al. 2001) with three options per block, and depends on a digital platform, ClimMob (Quirós et al. 2023). Tricot uses option ranking as a main way to elicit farmers' observations and preferences, but protocols created by crop breeding programs also involve metric harvest measurements. Therefore, the best way to obtain these measurements needs to be identified.

Obtaining yield estimates on smallholder farms is often complicated due to: (i) the need to use comparable units of measurements, (ii) differences in harvesting times within a cropping season, (iii) mixed cropping and irregular plot sizes, (iv) loss of plants within the cropping season, and (v) piecemeal harvesting (Fermont et al. 2009; Fermont and Benso 2011; Murphy et al. 1991; Reynolds et al. 2015). Another challenge of yield estimates by smallholder farmers is the human factor—the reliability of their estimates and their ability to collect and communicate data (Ochieng et al. 2019). Little is known about farmers' ability to produce accurate yield estimates. Therefore, agricultural researchers tend to avoid relying on farmers and work with technicians instead. This assumes, however, that there is no trade-off between measurement accuracy and other factors, such as trial scalability, the need to reach more remote locations to ensure trial representativeness, or

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

the relative cost of data points of technicians' versus farmers' measurements. On-farm trials are a deliberately designed socio-technical endeavor that respond to multiple needs and requirements; however, their design often fails to take into account their specific goals and characteristics (Kool et al. 2020). To make scientifically supported judgments on the relative merits of farmers' yield measurements, better insights into their abilities must be gained.

This study focuses on one aspect of accuracy of farmer yield estimates—the measurement of harvest output once this is obtained. We do not focus on area estimation, which is less relevant for trials in which the planting arrangement is standardized. Yields, in the context of on-farm trials, are determined with the *planted* area in the denominator and not the *harvested* area, which may be smaller (cf. Wahab 2020). Partial crop loss poses a challenge here. Also, we leave for future research two relevant issues that are directly relevant to harvest measurement: (i) the discrimination of the marketable product from overall biological output, and (ii) determining the moisture rate of the product.

We tested different methods for farmer yield assessments, both for their accuracy in estimating harvest output and for factors influencing this accuracy. We focused on maize (*Zea mays*) kernels and cocoa (*Theobroma cacao*) beans. We compared different methods against the simple benchmarks of ranking weights by lifting sacks and visual volume estimation. One hypothesis is that volumetric measurement using a container with a known volume provides the right trade-off between the inherent accuracy of the method itself and its proneness to human mistakes. Using a scale to weigh would theoretically be the most precise method. Nonetheless, as it is a more complex method and not all farmers may be familiar with it, mistakes may occur. Another hypothesis is that farmers can increase accuracy through training to use a weighing scale. This study determines the accuracy of farmer measurement outputs and determinants of the accuracy by using simple on-farm exercises in which farmers execute the estimation and measurement tasks. The results of this study provide researchers with important information about the type and the magnitude of error that they can expect when farmers measure harvest output.

2. Background

2.1. Previous studies

Here, we provide an overview of previous work assessing farmers' accuracy in measuring crop yields—in relation to on-farm experimentation and also agricultural surveys—which often rely on farmer-generated harvest estimates that we deem to be the most relevant.

Gay and Cole (1967) provide evidence from Liberia in the only study known to us in which volumetric estimation was evaluated. Based on a small sample, they conclude that farmers can provide highly accurate visual volumetric estimates of small rice volumes, in contrast with Peace

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Corps volunteers, who did a poor job. This suggests that farmers are able to execute measurement tasks in the context with which they are highly familiar.

Steinke et al. (2017) evaluated the accuracy of farmer-generated data in agricultural citizen science by asking Honduran farmers to rank common bean varieties for pest damage, disease severity, vigor, and architecture. Some traits were difficult to assess for farmers. The authors showed, however, that when the farmers' observations are aggregated across plots, they matched the rankings established by technicians. It is important to note that Steinke et al. (2017) studied only pre-harvest characteristics and did not focus on yield. The study is relevant, however, as we also focus on rankings as one way to collect data from farmers.

Closer to our own study, Ochieng et al. (2019) compared Kenyan farmers' yield measurements with researchers' measurements, working with common bean trials. Ochieng et al (2019) adopted the volumetric approach, where the farmers used a 250 ml tin to estimate yield, although they only considered volumetric harvest output measurement by farmers, but did not compare this to farmers' ranking or weighing harvest output. Similar to Steinke et al. (2017), Ochieng et al. (2019) found that individual farmer observations had relatively high variation, but that aggregated observations across varieties had a very high correlation with the researcher-produced data ($r=0.98$). Thus, conclusions drawn from volumetric data collected from farmers would be similar to conclusions drawn from metric data collected from researchers (Ibid).

A recent study was conducted in parallel with this one, measuring farmers' accuracy in estimating volumetric measurements of actual plot yields, using a more realistic setup than ours (Nabateregga et al. 2024). This study presented similar findings on volumetric measurements. Farmers produced reliable results, with an r^2 of 0.95 between farmers' yield estimates with tin cans and researcher-determined weights.

Surveys often employ yield estimation procedures and the literature about the accuracy of survey methods also provides relevant insights. Important methods for survey-yield estimates are crop cuts and farmer visual estimates (FAO 1982; Murphy et al. 1991; Casley and Kumar 1998). Both methods are described by Fermont and Benson (2011). The crop cut is based on sampling from subplots within the total plot size, using different sampling methods. Crop cuts are labor intensive, expensive, and time consuming. Farmer estimates can be obtained before and after harvesting, either before weighing or from farmers' measurements. Farmer estimates are much cheaper than crop cuts. For example, Kosmowski et al. (2020) used time measurements to estimate the cost of executing crop cuts in Ethiopia. In their study, the cost of crop cuts was 14–29 times higher than farmer yield reports.

Casley and Kumar (1998, p. 105–108) reported on several studies that showed an average of 14–18% upward bias for crop cuts, while farmer estimates showed less bias. Some of the

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

difficulties were related to sampling the field randomly and accounting for border effects. Murphy et al. (1991, p. 14–15) provide a list of seven possible biases in crop cuts, which are all in the upward direction. Empirically, upward biases for crop cuts have often been found. Verma et al. (1991) compared different yield estimation methods for five African countries. Crop cuts followed two sampling methods, measuring yield for parts of one row and measuring it for a square area of 5 x 5 m, and extrapolating the results to the entire plot. Farmers' estimates were obtained before the harvest and after harvesting, before weighing. The authors reported that crop cuts overestimated real yield by 34% for row samples (r=0.84) and 36% for squared samples (r=0.73) on average. Before the harvest, farmers tended to overestimate their harvest by 9% (r=0.83). After the harvest, having only visually assessed the harvest output, farmers overestimated production by 3% on average (r=0.88). The authors conclude that generally farmers' estimates are more accurate than crop cuts. It may be objected that the crop cuts used in this study were suboptimal, especially in how field sampling was conducted. In a recent study, Kosmowski et al. (2020) found significant differences between different sampling methods in crop-cut measurements. Their best crop-cut method (random octant) reduced the mean average error to 10.6 qt/ha. They compared this to the error of 23 qt/ha obtained by farmer visual pre-harvest estimates. However, this is comparing two different quantities, pre-harvest estimates and post-harvest observations. The authors fail to make the much more relevant comparison between researchers' and farmers' post-harvest output measurements, as farmer's post-harvest output measurements have a much smaller margin of error than their pre-harvest estimations (Verma et al. 1991; see the numbers shown above).

The studies provide insights into the order of magnitude of estimation errors, which are useful for comparison with our study. Given the high cost of crop cuts or total harvesting with field technicians, the use of farmer-measured harvest outputs is an important option. It may be combined with subsampling a portion of an on-farm trial for measurements by technicians. To gain insights into the relative value of farmer measurements, we need more precise knowledge of farmers' precision in measuring. The reported studies do not focus on farmer yield measurement, or base their results only on farmers' visual estimates before weighing. This study focuses on the other options available for farmer measurement of crop harvest output in on-farm trials.

2.2. Farmer measurement methods and their accuracy

In this study, we tested four ways to estimate harvest output: by lifting sacks, visual estimation of heaps, volumetric measurement with containers, and measurement using weighing scales. We are interested in the first two methods mainly as benchmarks for easy comparison with the other tests reported in the previous section, but also because measurements obtained from these simpler methods may be available in situations arising in trials due to farmers' decisions, loss of a measuring device, or turnover in technical staff or farmers who guide the experiments locally.

Each method produces different errors.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

- **Lifting sacks** and **visual estimation** are highly dependent on personal experience and ability. Jones (1986) provides a range of 3–12% for lifting weight discrimination that is right 75% of the time based on lifting. However, these studies generally focus on much smaller weights than those relevant for yield estimation.
- **Visual estimation** is accurate according to Verma et al. (1991) who reported that farmers on average overestimate by 3% and show a high correlation with real weight (r=0.88).
- **Volumetric measurement** can be affected by counting mistakes or differences in the filling of containers, such as filling to, beyond, or under the brim, or the degree of grain compaction.
- Measuring with a scale is potentially a highly accurate method, but many mistakes can still be made. Murphy et al. (1991) indicated that faulty weighing scales, inappropriate weighing scales, and the (non-)taring of containers could be sources of weighing bias – the latter being found to be the most common source of bias. Other errors can be reading errors, for example mistaking kilograms and pounds on scales that display both.

Several factors can affect farmers' ability to accurately determine outputs and weights, which we call *determinants of accuracy* here. A person's eyesight could play an influential role; a person with vision issues may not be able to read the scale accurately or see sufficiently well to visually estimate weight. Experienced farmers and farmers with more farming experience or longer participation in external projects may have been exposed to more opportunities to acquire skills that lead to more measurement accuracy compared to less-experienced farmers. Education can play a role in farmers' ability to accurately determine outputs and weights, for example, reading a scale and recording weights, which are skills that are often acquired through formal education. Gender can also influence people's ability to accurately determine weights and outputs, through other more proximal factors, especially familiarity with tasks that involve measuring weights or volumes. Interventions can counteract some of these determinants by providing training on measurement approaches, which could enhance farmers' accuracy in producing harvest output measurements.

2. Materials and Methods

2.1. Overview and general aspects

We selected five rural communities among the major cocoa and maize growing areas in Ghana. We focused our study on cocoa and maize as both crops are ubiquitous and easy to obtain in our study area and have products of contrasting sizes: dried beans for cocoa and shelled grains for maize.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

The different test methods employed in this study are hereinafter referred to as exercises. Each of the following four exercises were conducted for each of the two study crops (cocoa and maize) and in of the five study sites.

Exercise 1: Ranking weights based on lifting
Exercise 2: Visual volume ranking and weight estimation
Exercise 3: Volumetric measurement with a tin can
Exercise 4: Weighing using a scale

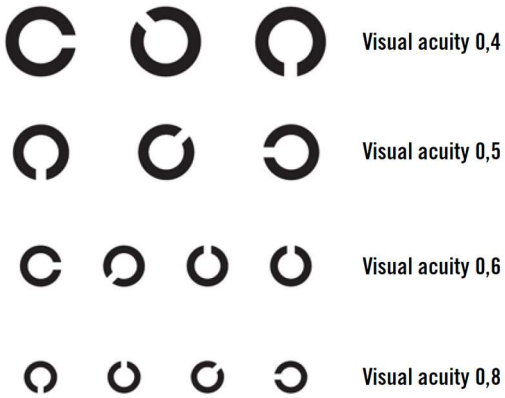
We presented farmers with relatively small quantities of crop products, as our direct interest was the estimation of harvest output from relatively small trial plots. Also, this helped to keep the exercises manageable. We completed a test round of the exercises and adjusted our exercise formats and questionnaires based on our test-round findings.

We were interested in the circumstances that influence accuracy. A questionnaire was administered to obtain information potentially associated with participants' abilities. We hypothesized that practical experience with a weighing scale and farming in general would exert an influence, as would the small differences in experimental setup between locations, such as daylight intensity. Also, we hypothesized that farmers' eyesight would influence their ability to perform the visual tasks (especially Exercise 2, but also Exercises 3 and 4). We asked farmers if they had eyesight problems and whether they habitually used glasses. Farmers' visual acuity was then tested by observing a Snellen chart, placed at a distance of 6 m from the farmers (Figure 1).

Regarding Exercise 4, we hypothesized that farmers would drastically increase their accuracy if they underwent a short training in using a butcher's scale. We tested this hypothesis by training some of the farmers and comparing their measuring results with those of the untrained farmers as a control. The approach is described in detail in section 2.6—Exercise 4.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

A



B



Figure 1. A. Snellen visual acuity chart; B. Executing the visual acuity test

Ethical clearance was obtained from the Alliance of Bioversity and CIAT's institutional ethical review board prior to the study (#2021-IRB29). Data presented here are not linked to personal data and cannot be traced to individual farmers. Approvals were sought from farmers for video recording, photography, and presenting their images in this manuscript.

2.2. Study locations and participants

The study took place in five rural locations within the major cocoa-growing areas of Ghana—the Western North and Ashanti regions (see Supplementary Figure S1). We worked with 40 farmers in five communities. We selected three communities in Western North region: Kojokrom (6.383 N, -2.600 W), Ched (6.118 N, -2.550 W), and Bopa 6.148 N, -2.567 W). We selected two communities in the Ashanti region: Addaikrom (6.700 N, -2.267 W) and Kojofordjourkrom (6.457 N, -2.167 W). Farmers in these communities predominantly grow cocoa, but also plant maize, cassava, plantain, and other food crops.

Kokoo Pa Farmer Association played a key role in the study. The association's agricultural extension agents (AEAs) mobilized its members by soliciting their interest and willingness to participate in the study. They approached farmers who were part of the ongoing tricot on-farm trials managed by Kokoo Pa. Trial participants all had access to land. The AEAs tried to ensure that both men and women, adults and youth participated. In all, 40 farmers participated in the study, all of whom had experience in growing both cocoa and maize.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Participating farmers' characteristics (gender, age, education level, participation in farmer schools, and previous experience with using butchers' scales) are presented in Table 1. Of the 40 farmers, only 6 had no formal education. Most of the farmers (excluding the 17–24 age cohort) had over 20 years' experience of growing cocoa and maize. Most of the farmers indicated that they have no eye-problem. Among the 7 farmers who have prior knowledge in butcher's scale, 3 of them used it to weigh cocoa beans while 4 used it to weigh nails.

Table 1. Selected characteristics of farmers involved in the study

Characteristics	Category	Participants N = 40
Gender	Male	25
	Female	15
Age	17–24	4
	35–40	7
	41–50	12
	51–60	12
	80	1
Education	No formal schooling	6
	Basic school	7
	Junior /middle school	22
	Secondary school	5
Participation in farmer schools	No	10
	Yes	30
Prior use of butchers' scales	No	33
	Yes	7

2.2. General aspects of the experimental setup

For each community, a single lot of maize grains and a single lot of cocoa beans were sourced from a farmer or cocoa purchasing clerk. This ensured that farmers observed and evaluated homogeneous lots of grains or beans in terms of moisture content and other product characteristics.

To avoid mutual influence and to ensure independent observations, farmers were kept 50 m away from where the exercises were set up and called individually to make their observations.

One of the goals of the study was to determine if training in using scales would lead to higher accuracy. Ten of the farmers in Addaikrom and ten in Kojofordjourkrom were trained on how to use weighing scales, following a ten-step procedure (Supporting materials 1). These 20 trained farmers' observations were compared with 20 untrained farmers' observations in the weighing test (Exercise 4 below).

The study employed different measurement devices, such as sacks for lifting and ranking, an empty tin can of 15 cm in diameter and 12 cm in height for volumetric measurement, and butchers' scales

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

for weighing (Figure 2A, 2B, and 2C, respectively). These measurement tools were selected to mimic what the farmers normally use in measuring their farm produce.

A B C



Figure 2. Measuring devices used for the measurement exercises. A. a 100 cm x 58 cm sack; B. an empty tin can; C. butchers' scales.

2.3. Exercise 1 – Ranking weights based on lifting sacks

Ranking weights based on lifting sacks containing different volumes of cocoa beans and maize grains is one way in which farmers can assess harvest output. Ranking is used for measuring several crop variety traits in the tricot approach (de Sousa et al. 2023). The same exercise setup was used in all the study locations. White sacks measuring 100 cm x 58 cm with five different weights of cocoa beans and maize were arranged randomly in a line on a flat surface for farmers to rank (Figure 3). One of each series of five sacks contained the reference weight, while the other sacks in each series differed from the reference weight by either +/-6% or +/-15% (Table 2). Exercise 1 used reference weights of 20 kg for cocoa beans and 25 kg for maize kernels. These sack weights were deemed sufficiently modest to allow women and elderly people to lift for the time needed to rank the sacks. Farmers were asked to lift each of the five sacks and rank each by saying which sacks they felt were heavier or lighter. Farmers were free to lift the sacks more than once. The farmers ranked the sacks in order of weight.

Table 2. Weight ranges used for farmers ranking sacks of cocoa beans and maize grains in Exercise 1.

Crop	**Much lighter** (-15%)	**Slightly Lighter** (-6%)	**Reference weight**	**Slightly Heavier** (+6%)	**Much Heavier** (+15%)
Maize	21.25 kg	23.5 kg	25 kg	26.5 kg	28.75 kg
Cocoa	17 kg	18.8 kg	20 kg	21.2 kg	23 kg

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions



Figure 3. Cocoa beans in sacks arranged for farmer ranking.

2.4. Exercise 2 - Visual volume ranking and weight estimation

This method is a benchmark as farmers visually predict the weights of their produce before it is weighed. Five heaps of known weights and bags of cocoa beans and maize grains were arranged separately on black polythene sheets (Figure 4; Table 1). For every location, flat, uniform surfaces were selected for the heaps. This exercise used reference weights of 64.3 kg cocoa beans and 25 kg of maize, with other heaps differed by 5% or 10% (Table 3). These percentages are lower than those for Exercise 1 as we expected that visual estimation would be more accurate than lifting. Farmers were requested to estimate how much these heaps translate into bags (standard bags commonly used by the farmers in the study area for cocoa and maize) and the actual weights. Farmers observed the heaps from different angles and made their predictions.

Table 3. Weight ranges used for farmers ranking the heaps of cocoa beans and maize grains used for Exercise 2.

Crop	Much lighter (-10%)	Slightly lighter (-5%)	Reference Weight	Slightly heavier (+5%)	Much heavier (+10%)
Maize	22.5 kg	23.75 kg	25 kg	26.25 kg	27.5 kg
Cocoa	57.87 kg	61.1 kg	64.3 kg	67.52 kg	70.73 kg

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

A B



Figure 4. Heaps arranged for visual weight estimation: A. maize; B. cocoa.

2.5. Exercise 3 - Volumetric measurement with tin can

A known weight of cocoa beans (64.3 kg) or maize grains (25 kg) was poured on black polythene sheets. Farmers were provided with a tomato tin can and requested to measure the volume of beans or grains by filling a sack (Figure 4). Farmers reported the number of tin cans for each of the volumes.



Figure 5. Farmer performing volumetric measurements with a tin can

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

2.6. Exercise 4 - Weighing using a scale

Farmers were requested to weigh the cocoa beans and maize grains using butcher's scales while the researcher observed and recorded the farmers' experience. Under this exercise, farmers were given a scale, a bowl, and sacks with cocoa beans (4 kg) and maize grains (2.5 kg). The weighing of the cocoa beans and maize grains were not done on the same day. The butcher's scale that was used displayed both pounds and kilograms (Figure 2C). Using a butcher's scale represents a feasible and cost-effective method. In contrast, traditional cocoa scales are bulky and are difficult to be transported around. The drawback is that only small amounts can be weighed at the same time, in contrast with the volumetric measurement. This can result in additional errors if the numbers are added up by hand. We will compare the values directly, however. This assumes no calculation mistakes are made and that the error distributions will remain the same after adding up the values, as farmers will tend to make the same mistakes when repeatedly weighing samples from the same plot. The same weighing scales, taring containers and product weights were used in all the communities to avoid weighing biases among farmers of different communities. The calibration of the butcher's scale is similar to that of the traditional cocoa weighing scales.

We trained farmers in Addaikrom and Kojofordjourkrom (n=20) in weighing cocoa using a scale, using a well-designed training manual (Ulzen and Van Etten 2021). The training booklet was supplied to the farmers during the training before the hands-on part of the training. Farmers were given hands-on training on the different steps of the process, including placing the scales on a flat surface, taring, calibration, recording of weighted values and keeping scale free of additional weight.

A. B.



Figure 6. A farmer weighing cocoa beans with a butcher's scale. A. Pouring samples into a bowl. B. observing the weight of the cocoa beans in the bowl.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

2.7. Statistical analysis

All the data generated in this study were analyzed in R software (R Core Team 2021). The R packages janitor (Firke 2021), ggplot2 (Wickham, 2016), tidymodels (Kuhn et al. 2020), and tidyverse (Wickham et al. 2019) were used in the analysis.

For exercises 1 and 2, we estimated the Just Noticeable Difference ($\Delta I_{0.75}$) as a comparable value. This is the weight difference at which the observers have 75% probability of being able to correctly detect a difference in weight. The 75% threshold, which is commonly used, represents 50% probability of correct ranking after correcting for random guessing. We calculated this using the expected z-score of the difference between measurements from two normal distributions:

$$Z_{0.75} = 0.6745 = \frac{\Delta I_{0.75}}{\sqrt{2\sigma^2}} \rightarrow \Delta I_{0.75} = 0.6745 \cdot \sqrt{2\sigma^2} \quad (1)$$

To obtain sigma values in kilograms, from the data obtained from Exercises 1 and 2 we used the Thurstonian (Case V) model (Thurstone 1927). This model assumes that the rankings provided by the farmers for the sacks are driven by latent variables that follow a normal distribution. From the Thurstonian model, we estimated the worth values parameters by constraining the standard deviation (sigma) to 1 (Thurstone 1927). We then used linear regression to relate the estimated worth values to the actual weights and derived sigma values in kilograms. These values were then used to calculate $\Delta I_{0.75}$ using Equation 1.

For exercises 2, 3 and 4, we estimated the deviations from the true value as the root of the mean squared deviance (RMSD), and the mean absolute deviance (MAD), and also expressed this as a percentage of the reference weight.

We used linear regression models to determine which circumstances predict the magnitude of the relative deviation in the different exercises.

3. Results

3.1. Accuracy

Table 4 shows the results in terms of the Just Noticeable Difference based on ranking only. Farmers discriminated correctly between a heavier and a lighter bag 75% of the times or more when the difference exceeded 8.5% (maize) or 6.3% (cocoa) of the reference weight. Visual ranking led to larger relative errors than lifting, discriminating 75% correctly at 6.4% (maize) and 14.6 % (cocoa)

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

of the reference weight. The results give us a benchmark to compare with more sophisticated measures.

Table 4. Reference weights and Just Noticeable Difference for exercises 1 and 2, based on rankings.

Exercise	Crop	Ref (kg)	$\Delta I_{0.75}$ (kg)	$\Delta I_{0.75}$ (%)
1a. Lifting ranking	Maize	25.0	2.13	8.5
1b. Lifting ranking	Cocoa	20.0	1.26	6.3
2a. Visual ranking	Maize	25.0	2.05	6.4
2b. Visual ranking	Cocoa	64.3	9.36	14.6

Table 5 shows the results of exercises 2, 3 and 4 for the deviations from the true value. Figure 8 shows the same results visually. For exercise 2, comparing Table 4 and 5 shows that the $\Delta I_{0.75}$ value is lower than the RMSD (%) value. This implies that farmers are relatively accurate in pairwise visual comparisons for weight, but they are less accurate in visually determining the absolute weight. Even so, farmers achieved remarkably low bias (MSD) in the visual estimation of maize and cocoa.

The tin can measurement achieved the lowest errors for all metrics, with one exception. In the case of cocoa, trained farmers achieved better results with the scale for MAD, but produced some large outliers leading to a higher RMSD and were more biased, overestimating the weight (high MSD). For tin can measurements, the bias was positive for maize and negative for cocoa.

The results for exercise 4 show very poor results for farmers without prior training, who did not improve (much) compared to the benchmark, the visual estimation of heap volumes. Training helped to improve the accuracy for all metrics for both crops. Despite this improvement, taken together the scale measurements fail to outperform the tin can measurements.

Table 5. Reference weights and deviance from true weight for exercises 2 to 4. RMSD, Root Mean Squared Deviation; MAD, Mean Absolute Deviation; MSD, Mean Signed Deviation

Exercise	Crop	Ref. (kg)	RMSD (kg)	RMSD (%)	MAD (kg)	MAD (%)	MSD (kg)	MSD (%)
2a. Visual estimation	Maize	25.0	4.53	18.1	3.27	13.1	-0.60	-2.4
2b. Visual estimation	Cocoa	64.3	10.05	15.6	7.04	11.0	-3.81	-5.9
3a. Tin can measurement	Maize	25.0	1.85	7.4	1.38	5.5	0.51	2.0
3b. Tin can measurement	Cocoa	64.3	4.29	6.7	3.26	5.1	-1.85	-2.9
4a. Scale - trained	Maize	2.5	0.47	18.8	0.31	12.2	0.28	11.0
4b. Scale - trained	Cocoa	4.0	0.30	7.6	0.19	4.8	0.17	4.3
4c. Scale - untrained	Maize	2.5	1.03	41.2	0.90	36.0	0.74	29.6
4d. Scale - untrained	Cocoa	4.0	0.52	13.1	0.44	10.9	0.36	8.9

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

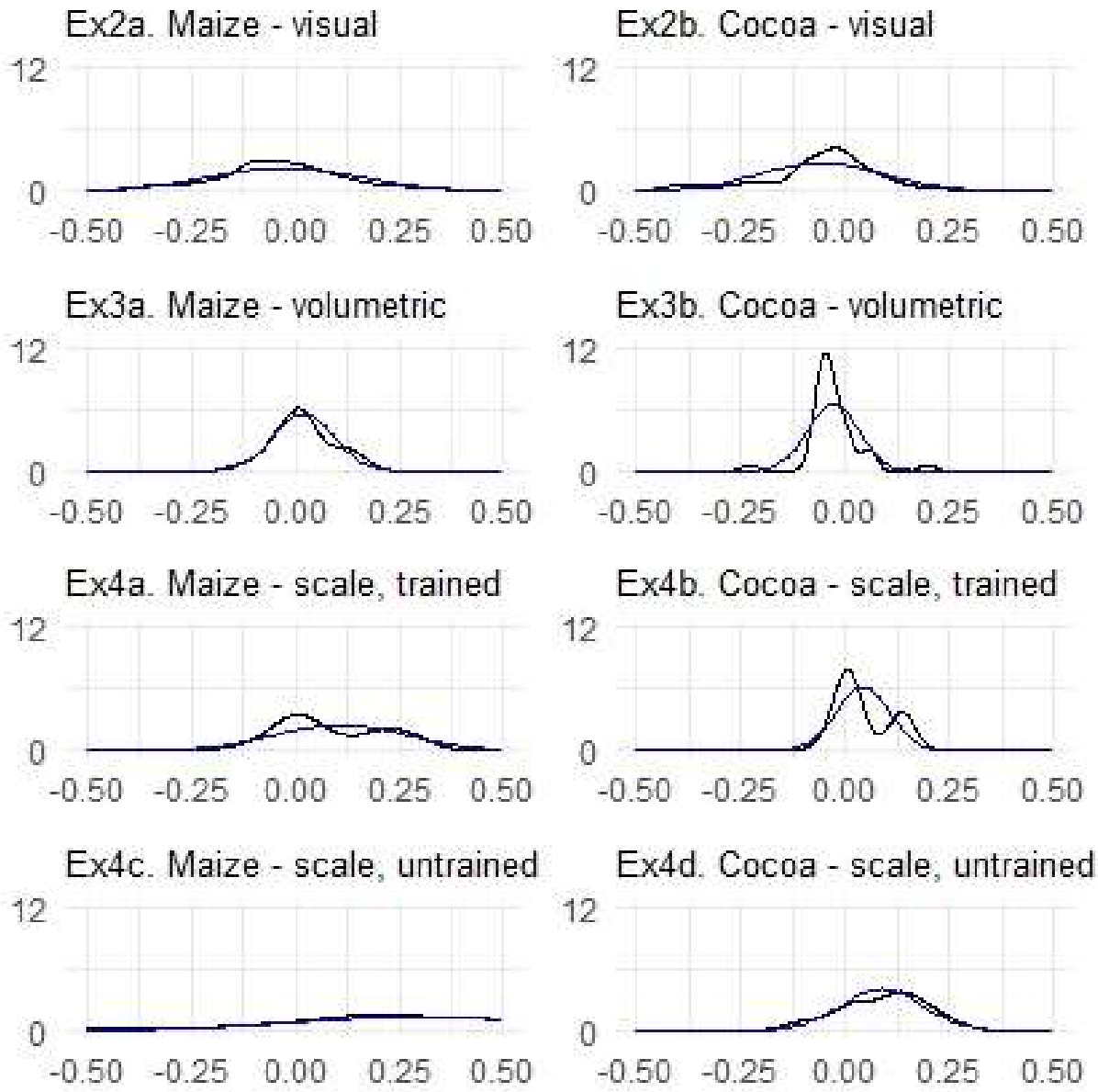


Figure 8. Distributions of relative errors from the true weight in Exercises 2 to 4. The black lines represent the density of observations, while the blue line represents the corresponding normal distribution. The relative error is calculated as (measured value – true value) / true value.

3.2. Exercises 3 and 4. Procedure execution

For the volumetric measurement of farmer outputs, an important source of error is the level of fullness of the can. Some farmers fill the can to the brim, whereas others fill the can beyond the brim. A few mistakes were made with recording the numbers as well.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

For the weighing exercises, we recorded detailed data on mistakes. Without training, 0% of the farmers executed all the steps correctly for either cocoa or maize (n=20 for each). With training, this improved to 55% of the farmers executing a correct weighing of cocoa, and 45% for maize. Many mistakes were made with reading out the number. Farmers obtained correct readings in only 10% of the cases before training and 57% the cases after training. One mistake was that farmers often confused kilograms and pounds, not following the instructions to read kilograms. Another source of errors was how farmers addressed taring and correcting for the weight of the container, despite training. Many farmers forgot to tare with containers and then forgot to subtract the weight of the container.

3.3. Determinants of farmer accuracy

We performed exploratory regression analyses for each exercise to relate farmers' characteristics to their error rates in estimating yield outputs. Interestingly, we found that farmers' accuracy in heap volume estimation (exercise 2) was affected by location, eyesight, education, and previous experience. Their accuracy using tin cans (exercise 3), however, was only affected by location, in one case. The same location effect is present for maize weighing. Farmers' accuracy in weighing (exercise 4) was affected by a range of factors, including eyesight, experience, education, location. These findings show that volumetric weighing using tin cans is the most robust to differences between farmers.

4. Discussion

4.1 Remaining limitations

A limitation of this study is that it did not consider moisture rates. Data on moisture content is needed to convert gross weights into dry matter yield, or grain yield with a given moisture rate (e.g. 14%). Previous studies found that different moisture levels affected farmers' yield estimates considerably (Casley and Kumar 1989; Murphy et al. 1991). Nabateregga et al. (2024) instructed farmers to dry common beans samples for at least two days and then measured moisture rates. The remaining variation in moisture rates had a significant variety effect, however. Given the variability in moisture between plots, this problem requires better field data on moisture ranges. Simple methods are needed to determine moisture rates.

Another issue is the discrimination of the marketable product from overall biological output. Some proportion of the product may not be saleable at the market due to its limited product quality. For some situations, this is an important issue. Farmers may be able to discriminate product quality classes well, but methods are needed to ensure consistency.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

4.2. Comparisons between exercises and with other studies

Weight comparisons based on lifting produced values for the Just Noticeable Difference in the range between 6.3 and 8.5% (Exercise 1). These values are within the 3-12% range provided by Jones (1986). These results show that comparing weights based on lifting bags will tend to result in low accuracy, but contain information.

The heap size visual estimations showed comparable levels of accuracy (Exercise 2). There is a downwards bias for cocoa, which may reflect weighing practices by buyers. The regression analysis for both maize and cocoa suggest that farmers' experience with the crop influences their accuracy. This corroborates the findings of Gay and Cole's (1967) study which showed that experienced farmers' were able to produce highly accurate volume estimates.

The probability of a right answer on which sample has more volume depends on the size of the difference. When differences are larger than 6.3-6.4%, this probability exceeds 75% using the best method for maize or cocoa. In that sense, estimation without instruments could provide additional verification for measurements, which is relevant for data quality control when samples from the same farmer show large differences.

Tin can measurements were the most accurate overall (Exercise 3). Also, we did not find differences in accuracy between farmers with different gender, education, eyesight, experience, or location in this sample. This suggests that this approach is highly robust. Remaining inaccuracy was mainly related to the manner in which the tin cans are filled. Farmers systematically pack more grains or fewer beans into it. This is done very consistently, especially for cocoa, as can be easily appreciated in Figure 8. Therefore, the method can be recalibrated to produce less biased results. Recalibration possibly produces less bias than training farmers to pack more beans into the cans, which may lead to some farmers doing it and other farmers forgetting it. Similarly, the bias of maize volumetric measurement can be reduced even further through recalibration. With correction of the calibration bias and better reporting on tin can fractions, our results suggest that the method could achieve a MAD of 2-3% or lower.

Weighing resulted in larger errors than tin can measurements, and for maize did not outperform visual estimation in percentage points. Here, an upward bias is a main contributor to the error. Not taring the container and reading pounds instead of kilograms both produced an upward bias. There is a large difference between maize and cocoa, with cocoa giving much higher accuracy. We found a few large mistakes for cocoa. Familiarity with cocoa weights may have helped farmers to correct themselves, especially when reading out an obviously wrong value.

One issue that Nabateregga et al. (2024) found was related to measuring small amounts. Near-zero yields were often rounded to zero, causing a downwards bias. As we did not include very low weights in this study, we did not find this bias in our data. Nabateregga et al. (2024) also found

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

that reporting of portions of cans induced granularity error, due to rounding. Tin can measurements were based on fairly large amounts of 20-25 kg in our study, leading to relatively accurate results.

4.3. Implications for volumetric measurement

The results of this study suggest that volumetric measurement is a robust approach that produces relatively accurate data. Its main strength is that it can be readily executed by farmers without any prior training. Its accuracy was not notably influenced by differences between farmers, which is an important indicator of its robustness in groups with low education levels. In our group, 27 out of 40 farmers had finished at least junior school, and only 6 had no education. In other contexts, farmers may have lower levels of education. In designing robust approaches for on-farm testing that will work in many contexts, volumetric measurements will have an important role to play.

The method produced some errors, especially with fractions. This error can potentially be reduced in various ways. Tin can size should be chosen carefully, in proportion to the expected yields. For small plots, this means that small tin cans should be provided to farmers. Another approach is to draw lines for fractions in the interior of tin cans. A third approach is to provide a combination of a large and a small tin can. The last bits are then measured in terms of the smaller tin cans, and farmers report two numbers separately. This can also cause new mistakes, such as mixing up the two numbers.

This method produces volume estimates, but volumes need to be converted to weights. This requires calibration. If seed sizes are very different, there may be variety effects. Measuring a small subset of samples with a scale would be a feasible strategy to produce calibration data.

4.4. Implications for weighing

Measurements using scales can theoretically lead to the most accurate results, but this study showed that this is difficult to achieve in practice. Using conventional butchers' scales induces many errors, leading to low accuracy compared to tin can measurements. Training improved the accuracy to a large degree and largely removed differences between farmers, but still did not achieve the level of accuracy of volumetric measurement.

This study also suggests that weighing with scales is not necessarily the best option. Implementation of this approach requires special care. The study has shown the clear need for farmers' training. Despite recent training, errors persisted. As measurements will usually take place several months after training, our results may be optimistic. Researchers will need to evaluate farmers' abilities to implement this approach, as some farmers may need help from others. Also, more effort in effective training methods may be needed.

Also, modifications to the procedure can help to overcome some of the identified problems. Reading errors can be further reduced by selecting scales with a single unit on the display

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

(kilograms or pounds), and by using electronic scales, which are easier to read than analogue ones. Reading errors can potentially be reduced by further training as well. One important problem we identified was with taring the container on butchers' scales. For coarse products such as cassava, weighing for on-farm trials has been done using a spring scale and a sack, causing negligible tare bias, as the sack has a very low weight compared to the product (B. Teeken *pers. comm.*). In Nigeria, this work involved lead farmers, who support around ten other farmers and are trained in data collection. It requires providing farmers with spring scales, which are relatively inexpensive. Using sacks and spring scales could similarly help to increase accuracy for grain crops and cocoa. One possible issue is the decalibration of spring scales and their loss in the field. A regular supply of spring scales and repeated training in their use should be factored in, if this approach is used.

4.5. General considerations

Volumetric measurement with tin cans consistently outperformed our benchmark, simple visual estimation of heaps. This approach required no specific training and was robust across different groups of farmers. Remarkably, weighing did not consistently outperform visual estimation. Therefore, visual estimations could play a role as a check in the approach. If farmers estimate the heap sizes before doing volumetric measurements, they could guard against large outliers, either by comparing with their own heap size estimates or by reporting both their initial estimate and the more precise number of tin cans.

Decisions on methodology should involve a consideration of the purpose of trials, costs, and practical limitations. There are trade-offs between the accuracy of a particular measurement and the size of the experiment. At present, trial managers often default to a reliance on technicians to visit fields and measure yields. Technicians' field visits are a main cost driver in on-farm trials (Kosmowski et al. 2020). Therefore, this will generally mean that field visits limit the resources that could be dedicated to increasing the number of locations covered by the on-farm trial. Also, on-farm trials managed with heavy involvement from technicians are often located in well-reachable places and may therefore induce geographical biases, potentially reducing external validity. Our results suggest that, given proper training and experimental design, a trial that relies to a large degree on farmers' yield measurements may provide more information value per dollar invested than a much smaller trial in which yields are exclusively measured by technicians. Tin cans can provide a cheap and robust method, while weighing is relatively more expensive and requires careful planning. Our study contributes new information to help trial designers in making more informed decisions, carefully considering the trade-offs implied by different on-farm testing strategies.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Acknowledgments

We are grateful to Kookoo Pa members who participated in this study. We acknowledge Olga Spellman (Alliance of Bioversity International and CIAT Science Writing Service) for copy editing the manuscript.

Financial support

The field work for this research was done under a project funded by NWO-WOTRO, W.08.240.109 "Using citizen science to improve agro-ecological and climatic accuracy of cacao planting material recommendations in Ghana and its regional-level access and delivery to farmers". Data analysis was supported by a project funded by the Bill and Melinda Gates Foundation, INV-031561, "1000FARMS Research Platform".

Competing interests

Competing interests: The authors declare none.

Authorship

Jacob Ulzen designed the research, led the data collection, analyzed the data, and wrote the manuscript. Nana Oduro Osei - Bonsu and George Bush collected the data. Hugo Andrés Dorado Betancourt analyzed the data. Jacob van Etten acquired funding, designed the research, analyzed the data, and wrote the manuscript. All authors reviewed and approved the manuscript.

Data availability statement

Data are available from Dataverse under the CC BY 4.0 license. https://creativecommons.org/licenses/by/4.0/
https://doi.org/10.7910/DVN/6VTLGD

References

Atlin, G.N., Cooper, M. and Bjørnstad, Å. (2001) A comparison of formal and participatory breeding approaches using selection theory. *Euphytica* 122, 463-475. https://doi.org/10.1023/A:1017557307800

Casley, D.J. and Kumar, K. (1988) *The Collection, Analysis and Use of Monitoring and Evaluation Data.* Baltimore, MD: Johns Hopkins University Press for the World Bank.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

van Etten, J. (2011) Crowdsourcing crop improvement in Sub-Saharan Africa: a proposal for a scalable and inclusive approach to food security. *IDS Bulletin* 42(4), 102-110. https://doi.org/10.1111/j.1759-5436.2011.00240.x.

FAO (Food and Agriculture Organization of the United Nations) (1982) *The Estimation of Crop Areas and Yields in Agricultural Statistics.* Economic and Social Development Paper No. 22. Rome, Italy: FAO.

Fermont, A.M., Van Asten, P.J., Tittonell, P., van Wijk, M.T. and Giller, K.E. (2009) Closing the cassava yield gap: an analysis from smallholder farms in East Africa. *Field Crops Research* 112(1), 24-36. https://doi.org/10.1016/j.fcr.2009.01.009.

Fermont, A. and Benson, T. (2011) *Estimating Yield of Food Crops Grown by Smallholder Farmers.* Washington DC: International Food Policy Research Institute.

Firke, S. (2021) janitor: Simple Tools for Examining and Cleaning Dirty Data. R package version 2.1.0. https://CRAN.R-project.org/package=janitor.

Gay, J. and Cole, M. (1967) *The New Mathematics and an Old Culture: A Study of Learning among the Kpelle of Liberia.* London: Holt, Rinehart and Winston.

Jones, L.A. (1986) Perception of force and weight: theory and research. *Psychological Bulletin* 100(1), 29-42. https://psycnet.apa.org/doi/10.1037/0033-2909.100.1.29

Kool, H., Andersson, J.A. and Giller, K.E. (2020) Reproducibility and external validity of on-farm experimental research in Africa. *Experimental Agriculture* 56(4), 587-607. https://doi.org/10.1017/S0014479720000174.

Lacoste, M., Cook, S., McNee, M., Gale, D., Ingram, J., Bellon-Maurel, V., MacMillan, T., Sylvester-Bradley, R., Kindred, D., Bramley, R. and Tremblay, N. (2022) On-farm experimentation to transform global agriculture. *Nature Food* 3(1), 11-18. https://doi.org/10.1038/s43016-021-00424-4.

Murphy, J., Casley, D.J. and Curry, J.J. (1991) *Farmers' Estimations as a Source of Production Data.* World Bank Technical Paper 132. Washington, DC: World Bank.

Nabateregga, M., Dorado, H., Solberg, S.Ø., van Etten, J., Gregory, T. and de Sousa, K. (2024) Accuracy of farmer-generated yield estimations of common bean in decentralised on-farm trials. *SSRN Preprint*. http://dx.doi.org/10.2139/ssrn.4863026.

Ochieng, H., Ojiem, J. and Otieno, J. (2019) Farmer versus researcher data collection methodologies: Understanding variations and associated trade-offs. *AfricArXiv Preprints.* https://doi.org/10.31730/osf.io/ncw8a.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Quirós, C., de Sousa, K., Steinke, J., Madriz, B., Laporte, M.A., Arnaud, E., Manners, R., Ortiz-Crespo, B., Müller, A. and van Etten, J. (2024) ClimMob: Software to support experimental citizen science in agriculture. *Computers and Electronics in Agriculture* 217, p. 108539. https://doi.org/10.1016/j.compag.2023.108539

R Core Team (2021) *R: A Language and Environment for Statistical Computing.* R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.

Reynolds, T.W., Anderson, C.L., Slakie, E. and Gugerty, M.K. (2015) How common crop yield measures misrepresent productivity among smallholder farmers. *2015 Conference, August 9-14, 2015, Milan, Italy* 212294, International Association of Agricultural Economists.

de Sousa, K., van Etten, J., Manners, R., et al. (2024) The tricot approach: an agile framework for decentralized on-farm testing supported by citizen science. A retrospective. *Agronomy for Sustainable Development* 44, 8. https://doi.org/10.1007/s13593-023-00937-1.

Steinke, J., van Etten, J. and Zelan, P.M. (2017) The accuracy of farmer-generated data in an agricultural citizen science methodology. *Agronomy for Sustainable Development* 37(4), 32. https://doi.org/10.1007/s13593-017-0441-y.

Thurstone, L. (1927) A Law of Comparative Judgment. *Psychological Review* 34, 273–286.

Verma, V., Marchant, T. and Scott, C. (1988) *Evaluation of Crop-Cut Methods and Farmer Reports for Estimating Crop Production: Results of a Methodological Study in Five African Countries.* London: Longacre Agricultural Development Centre Ltd. https://doi.org/10.22004/ag.econ.257587.

Wahab, I. (2020) In-season plot area loss and implications for yield estimation in smallholder rainfed farming systems at the village level in Sub-Saharan Africa. *GeoJournal* 85, 1553–1572. https://doi.org/10.1007/s10708-019-10039-9.

Waters-Bayer, A., Kristjanson, P., Wettasinha, C., van Veldhuizen, L., Quiroga, G., Swaans, K. and Douthwaite, B. (2015) Exploring the impact of farmer-led research supported by civil society organizations. *Agriculture & Food Security* 4, 1-7. https://doi.org/10.1186/s40066-015-0023-7.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L.D.A., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T.L., Miller, E., Bache, S.M., Müller, K., Ooms, J., Robinson, D., Seidel, D.P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K. and Yutani, H. (2019) Welcome to the Tidyverse. *Journal of Open Source Software* 4(43), 1686. https://doi.org/10.21105/joss.01686.

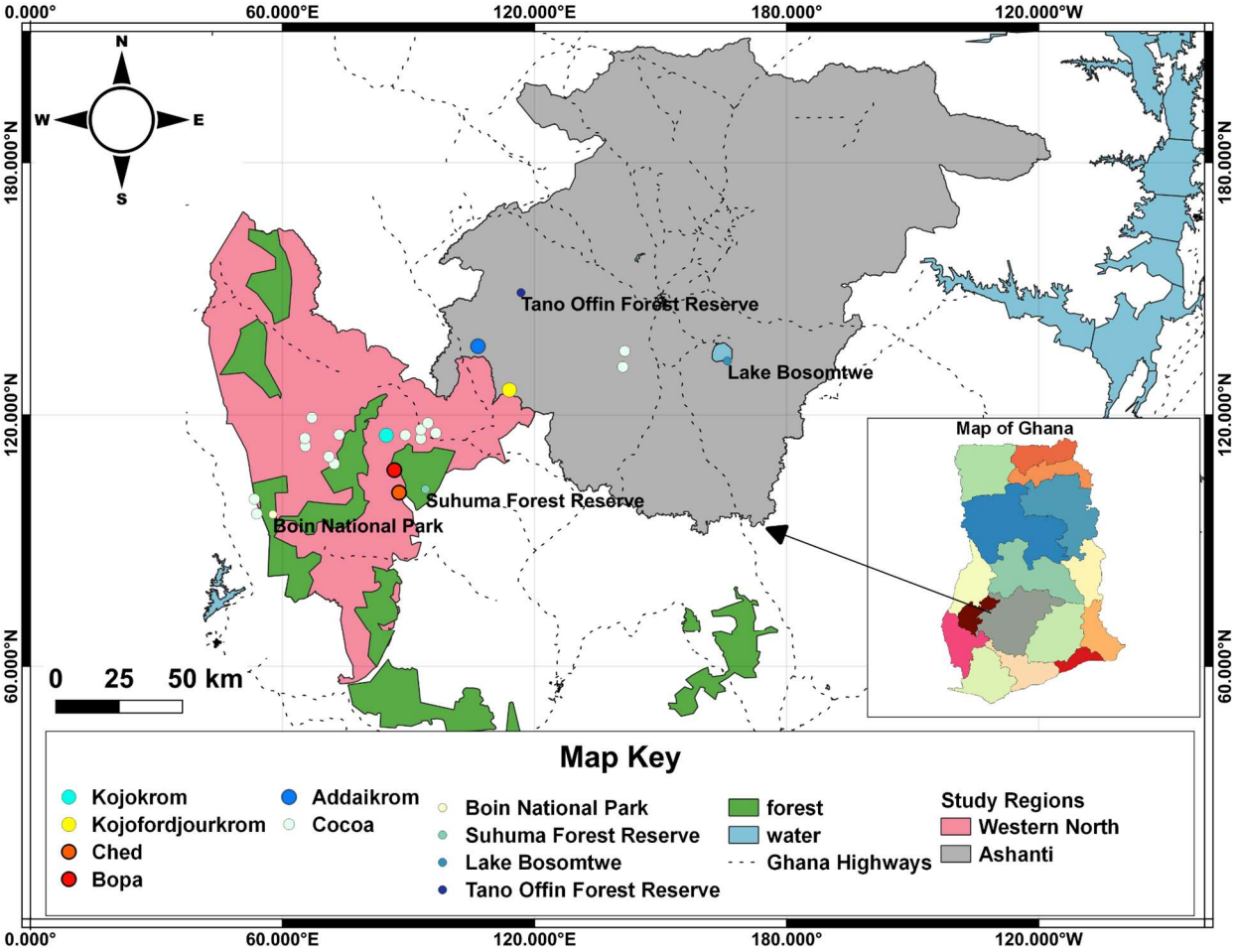
Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

1 Williamson, H.F. and Leonelli, S. (2022) Accelerating agriculture: Data-intensive plant breeding
2 and the use of genetic gain as an indicator for agricultural research and development. *Studies in*
3 *History and Philosophy of Science* 95, 167-176. https://doi.org/10.1016/j.shpsa.2022.08.006.

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Supplementary materials pertaining to Ulzen et al. Accuracy in agricultural citizen science: farmers' assessment of harvest output in Ghana

Figure S1. Map of Ghana indicating study locations



Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Tables S1 to S8 contain the results of regression analyses that explain farmers' accuracy (relative error) with their socio-economic characteristics (see Table 2 in the main text).

Table S1. Exercise 2 - visual estimation, cocoa. Socio-economic factors that influence relative error of farmer harvest output

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.11	0.007	15.901	**<0.001**
Age	0.006	0.014	0.439	0.66
Acuity_test	-0.022	0.013	-1.661	0.098
Location_Bopa	-0.056	0.014	-3.877	**0.00014**
Location_Ched	-0.027	0.014	-1.979	**0.049**
Location_Kojofordjourkrom	-0.047	0.016	-2.995	**0.0031**
Location_Kojokrom	-0.078	0.012	-6.637	**<0.001**
Education_Junior.high.Middle.School	-0.015	0.021	-0.721	0.47
Education_No.formal.education	0.01	0.015	0.671	0.50
Education_Secondary	-0.019	0.019	-1.005	0.32
Read_and_write_Yes	0.026	0.015	1.768	0.078
Gender_Male	0.003	0.01	0.306	0.76
Years_of_experience_between.10.20	-0.024	0.012	-1.978	**0.049**
Years_of_experience_between.3.5	-0.019	0.014	-1.398	0.16
Years_of_experience_between.5.10	-0.031	0.012	-2.656	**0.0086**
Participation_external_projects_Yes	-0.004	0.015	-0.241	0.810
Participation_farmer_schools_Yes	-0.005	0.015	-0.311	0.756
Under_what_circumstances_To.weigh.cocoa	-0.011	0.011	-1.051	0.29
Under_what_circumstances_Weighing.nails	0.012	0.011	1.136	0.25
Type_scale_use_None	-0.041	0.031	-1.313	0.19
Type_scale_use_Traditional.cocoa.weighing.scales	-0.055	0.03	-1.837	0.067
Eye_problem_Yes	-0.004	0.013	-0.327	0.74
Use_glasses_Yes	-0.01	0.011	-0.961	0.33
R squared	0.31			
Adjusted R-squared	0.23			
AIC	-340.34			
P-value	<0.001			

N = 195; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Table S2 Exercise 2 - visual estimation, maize. Socio-economic factors that influence relative error of farmer harvest output

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.131	0.007	18.343	**<0.001**
Age	0.017	0.016	1.051	0.29
Acuity_test	-0.011	0.012	-0.901	0.37
Location_Bopa	-0.076	0.016	-4.709	**<0.01**
Location_Ched	-0.012	0.013	-0.868	0.39
Location_Kojofordjourkrom	-0.066	0.019	-3.545	**0.0005**
Location_Kojokrom	-0.015	0.012	-1.238	0.22
Education_Junior.high.Middle.School	-0.098	0.021	-4.643	**<0.01**
Education_No.formal.education	-0.079	0.016	-5.11	**<0.01**
Education_Secondary	-0.118	0.02	-6.001	**<0.01**
Read_and_write_Yes	0.034	0.014	2.46	**0.015**
Gender_Male	0.001	0.011	0.091	0.93
Years_of_experience_between.1.3	0.045	0.011	4.214	**0.00004**
Years_of_experience_between.10.20	-0.033	0.013	-2.662	**0.0085**
Years_of_experience_between.3.5	0.015	0.013	1.183	0.24
Years_of_experience_between.5.10	-0.013	0.01	-1.275	0.20
Participation_external_projects_Yes	0.067	0.017	3.982	**0.0001**
Participation_farmer_schools_Yes	-0.051	0.015	-3.526	**0.0005**
Under_what_circumstances_To.weigh.cocoa	0.034	0.011	2.982	**0.0033**
Under_what_circumstances_Weighing.nails	0.011	0.009	1.19	0.24
Type_scale_use_None	-0.018	0.027	-0.648	0.52
Type_scale_use_Traditional.cocoa.weighing.scales	0.01	0.027	0.367	0.71
Eye_problem_Yes	-0.036	0.013	-2.704	**0.0075**
Use_glasses_Yes	0.011	0.011	0.973	0.33
R squared	0.44			
Adjusted R-squared	0.37			
AIC	-320.55			
P-value	<0.01			

N = 195; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Table S3 Exercise 3 - volumetric measurement, cocoa. Socio-economic factors that influence relative error of farmer harvest output

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.191	0.007	26.51	**<0.001**
Age	0.025	0.015	1.698	0.11
Acuity_test	0.014	0.014	1.025	0.32
Location_Bopa	-0.004	0.015	-0.285	0.78
Location_Ched	0.007	0.015	0.457	0.65
Location_Kojofordjourkrom	-0.01	0.016	-0.581	0.57
Location_Kojokrom	0.003	0.012	0.234	0.817
Education_Junior.high.Middle.School	0.005	0.022	0.206	0.84
Education_No.formal.education	-0.001	0.015	-0.043	0.97
Education_Secondary	0.024	0.02	1.192	0.25
Read_and_write_Yes	0.003	0.016	0.204	0.84
Gender_Male	-0.004	0.011	-0.398	0.70
Years_of_experience_between.10.20	0.014	0.013	1.078	0.30
Years_of_experience_between.3.5	0	0.014	-0.024	0.98
Years_of_experience_between.5.10	0.012	0.012	0.946	0.35
Participation_external_projects_Yes	0.003	0.016	0.208	0.84
Participation_farmer_schools_Yes	0.005	0.016	0.316	0.76
Under_what_circumstances_To.weigh.cocoa	-0.014	0.011	-1.213	0.24
Under_what_circumstances_Weighing.nails	-0.01	0.011	-0.877	0.39
Type_scale_use_None	-0.006	0.033	-0.194	0.85
Type_scale_use_Traditional.cocoa.weighing.scales	-0.002	0.032	-0.06	0.95
Eye_problem_Yes	0.016	0.013	1.182	0.25
Use_glasses_Yes	-0.02	0.011	-1.739	0.10
R squared	0.65			
Adjusted R-squared	0.20			
AIC	-119.94			
P-value	0.227			

N=40; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Table S4 Exercise 3 - volumetric measurement, maize. Socio-economic factors that influence relative error of farmer harvest output

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.055	0.007	7.50	**0.0000018**
Age	0.007	0.018	0.402	0.693
Acuity test	-0.004	0.013	-0.299	0.768
Location_Bopa	0.0038	0.0159	0.239	0.814
Location_Ched	0.002	0.016	0.147	0.884
Location_Kojofordjourkrom	0.044	0.018	2.415	**0.028**
Location_Kojokrom	0.0152	0.013	1.151	0.267
Education_Junior.high.Middle.School	-0.005	0.023	-0.222	0.827
Education_No.formal.education	0.0039	0.018	0.214	0.832
Education_Secondary	0.011	0.021	0.519	0.611
Read and write_Yes	0.0063	0.014	0.455	0.655
Gender_Male	-0.0009	0.0122	-0.078	0.938
Years_of_experience_between.1.3	-0.01	0.012	-0.810	0.430
Years_of_experience_between.10.20	0.0058	0.012	0.45	0.653
Years_of_experience_between.3.5	-0.014	0.0161	-0.877	0.394
Years_of_experience_between.5.10	0.0053	0.011	0.477	0.64
Participation_external_projects_Yes	-0.0037	0.017	-0.222	0.82
Participation_farmer_schools_Yes	-0.0092	0.02	-0.45	0.655
Under_what_circumstances_To.weigh.cocoa	0.0041	0.012	0.337	0.741
Under_what_circumstances_Weighing.nails	-0.0055	0.0119	-0.463	0.650
Type_scale_use_None	-0.027	0.033	-0.800	0.436
Type_scale_use_Traditional.cocoa.weighing.scales	-0.0138	0.0317	-0.434	0.671
Eye_problem_Yes	-0.0002	0.0141	-0.018	0.985
Use_glasses_Yes	0.0003	0.011	-0.026	0.979
R squared	0.67			
Adjusted R-squared	0.16			
AIC	-116.93			
P-value	0.297			

N = 39; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Table S5 Exercise 4, cocoa - trained. Factors that influence relative error of weighing output on cocoa of trained farmers

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.048	0.005	10.226	**0.000018**
Years of experience	0.002	0.013	0.162	0.88
Education	-0.027	0.010	-2.827	**0.026**
Age	-0.034	0.013	-2.634	**0.034**
Acuity test	0.028	0.011	2.456	**0.044**
Location_Kojofordjourkrom	-0.015	0.010	-1.490	0.180
Read_and_write (Yes)	-0.085	0.009	-9.044	**0.000041**
Gender_Male	-0.005	0.006	-0.876	0.41
Participation_external_projects_Yes	-0.040	0.010	-3.857	0.0062
Participation_farmer_schools_Yes	0.046	0.010	4.368	**0.0033**
Under_what_circumstances_To.weigh.cocoa	0.009	0.008	1.089	0.31
Eye_problem_Yes	-0.065	0.009	-7.392	**0.0002**
Use_glasses_Yes	0.058	0.008	7.384	**0.0002**
R squared	0.96			
Adjusted R-squared	0.88			
AIC	-91.09			
P-value	0.0013			

N = 20; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Table S6 Exercise 4, cocoa - untrained. Factors that influence relative error of weighing output on cocoa of trained farmers

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.109	0.008	13.598	**0.00004**
Years of experience	-0.017	0.020	-0.824	0.45
Education	0.019	0.032	0.603	0.57
Type of scale use	-0.017	0.021	-0.832	0.44
Age	0.033	0.014	2.397	0.0619
Acuity test	0.016	0.016	1.001	0.36
Location_Ched	-0.048	0.035	-1.382	0.23
Location_Kojokrom	-0.088	0.013	-6.643	**0.0012**
Read_and_write_Yes	0.023	0.048	0.472	0.66
Gender_male	0.006	0.019	0.315	0.77
Participation_external_projects_Yes	-0.010	0.013	-0.747	0.49
Participation_farmer_schools_Yes	-0.012	0.022	-0.532	0.62
Under_what_circumstances_Weighing_nails	-0.012	0.014	-0.871	0.42
Eye_problem_Yes	0.032	0.014	2.321	0.068
Use_glasses_Yes	-0.053	0.014	-3.701	**0.014**
R squared	0.93			
Adjusted R-squared	0.77			
AIC	-72.11			
P-value	0.034			

N = 20; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Table S7 Exercise 4, maize - trained. Factors that influence relative error of weighing output on cocoa of trained farmers

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.122	0.025	4.834	**0.002**
Years of experience	0.022	0.069	0.323	0.756
Education	-0.002	0.049	-0.049	0.963
Age	-0.047	0.066	-0.716	0.497
Acuity test	-0.030	0.061	-0.496	0.635
Location_Kojofordjourkrom	0.050	0.056	0.903	0.396
Read_and_write_Yes	-0.102	0.052	-1.951	0.092
Gender_Male	-0.068	0.031	-2.194	0.064
Participation_external_projects_Yes	-0.035	0.057	-0.621	0.554
Participation_farmer_schools_Yes	0.046	0.058	0.797	0.452
Under_what_circumstances_To.weigh.cocoa	-0.001	0.045	-0.029	0.978
Eye_problem_Yes	-0.104	0.048	-2.163	0.067
Use_glasses_Yes	0.056	0.043	1.311	0.231
R squared	0.78			
Adjusted R-squared	0.41			
AIC	-23.43			
P-value	0.17			

N = 20; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Table S8 Exercise 4, maize - untrained. Factors that influence relative error of weighing output on cocoa of trained farmers

Term	Estimate	Std.error	Statistic	P-value
(Intercept)	0.00	0.01	24.52	**0.000002**
Years of experience	0.01	0.03	0.43	0.6850
Education	0.06	0.05	1.18	0.2924
Type of scale use	-0.08	0.04	-2.17	0.0818
Age	-0.07	0.03	-2.58	**0.0493**
Acuity test	0.03	0.03	0.92	0.3981
Location_Ched	0.09	0.05	1.70	0.1495
Location_Kojokrom	0.23	0.03	8.79	**0.0003**
Read_and_write_Yes	0.04	0.07	0.61	0.5693
Gender_Male	0.02	0.03	0.44	0.6785
Participation_external_projects_Yes	0.02	0.02	0.70	0.5138
Participation_farmer_schools_Yes	-0.06	0.03	-2.04	0.0974
Under_what_circumstances_Weighing_nails	-0.07	0.03	-2.78	**0.0389**
Eye_problem_Yes	0.08	0.02	3.59	**0.0157**
Use_glasses_Yes	-0.0007	0.03	-0.03	0.9798
R squared	0.97			
Adjusted R-squared	0.90			
AIC	-47.88			
P-value	0.0054			

N = 20; Std = Standard

Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

R script

This script was used to produce the tables and figures.

```
# Preparations ------------------------------------------------
setwd(dirname(rstudioapi::getActiveDocumentContext()$path))

# Packages for data analysis
pacman::p_load(
  readxl, MASS, tidyverse, janitor, tidymodels, ggplot2,
  penalized, DescTools, lpSolve, irr, gosset, patchwork,
  mvtnorm, gtools, PlackettLuce
)

# Functions

# Create a contrast matrix to be used within the Thurstonian model
contrast.diff.mat <- function(len){
  c0 <- matrix(c(1,rep(0,len-2)),ncol=1)
  dM <- cbind(c0,matrix(0,ncol = len-1,nrow = len-1,byrow=T))

  for(i in 2:(len-1)){
    cs <- dM[,i-1]
    cs[cs==-1] <-0
    cs <- cs*-1
    cs[i] <-1
    dM[,i] <- cs

  }

  cf <- matrix(c(rep(0,len-2),-1),ncol=1)
  dM[,ncol(dM)] <- cf
  return(dM)
}

# Loglikehood function of the Thurstonian model
thurstonian_loglikehood <- function(alpha_bet){

  ln <- length(alpha_bet)
  C <- contrast.diff.mat(ln)
  mu <- as.matrix(alpha_bet)
```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
 1
 2    Sm <- C %*% diag(length(alpha_bet)) %*% t(C) # here the sigma is included, we are constraint
 3   to the identify matrix
 4    D <- diag(diag(Sm)^(-1/2))
 5    mean <- D%*% C %*% mu
 6    Smf <- D%*%Sm%*%D
 7    pmv <- pmvnorm(upper=mean[,1],lower=rep(-Inf,ln-1),mean=rep(0,ln-1),sigma= Smf) # First
 8   derivatives regard x and then y
 9    return(log(pmv[1]))
10   }
11
12   # Calculate Just Noticeable Difference
13   calcJND <- function(x, varnames, conv=FALSE){
14
15    if(conv) {x[,varnames[1]] <- 1/x[,varnames[1]] + runif(dim(x)[1])/1000}
16    x1 <- matrix(NA, ncol=length(unlist(unique(x[,varnames[2]]))), nrow=max(x[,varnames[3]]))
17    x1[as.matrix(x[,varnames[c(3:2)]])] <- unlist(x[,varnames[1]])
18    worth_x <- thurstone(as.rankings(x1))
19    dd <- as.data.frame(cbind(worth_x, weight=tapply(unlist(x[,varnames[4]]),
20   unlist(x[,varnames[2]]), mean)))
21    lm_x <- lm(weight ~ worth_x, data=dd)
22    result <- 0.6745 * sqrt(2*(coef(lm_x)[2]^2))
23    return(result)
24
25   }
26
27   calcStats <- function(x, y, id){
28    r <- rep(NA, times=3)
29    names(r) <- c('RMSD_true','MAD_true', 'MSD_true')
30    r['RMSD_true'] <- sqrt(mean((x-y)^2))
31    r['MAD_true'] <- mean(abs(x-y))
32    r['MSD_true'] <- mean(x-y)
33    return(r)
34   }
35
36   # Dataset processing
37   tricotNLLMU0 <- function(x, par){
38
39    par <- c(mu1=0,par)
40    # Process as a data.frame
```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
  matrixTricot <- as.data.frame(as.matrix(x))
  nam.par <- names(par)
  colnames(matrixTricot) <- nam.par

  # Initial values
  muMat <- data.frame(nam.mu=nam.par, par=par)

  # Create worth values matrix
  matrixTricot <- data.frame(Ind=row.names(matrixTricot),matrixTricot)
  matrixTricot <- matrixTricot %>% pivot_longer(cols = !Ind, names_to = "Item" ,values_to =
"Ranking")
  matrixTricot <- matrixTricot[mixedorder(matrixTricot$Ind),]
  matrixTricot <- matrixTricot %>% filter(Ranking!=0)
  matrixTricot <- left_join(matrixTricot,muMat,by=c('Item'='nam.mu'))
  matrixTricot <- matrixTricot %>% pivot_wider(id_cols = !Item,names_from =
Ranking,values_from = par,)
  matrixTricot <- matrixTricot[,-1]
  matrixTricot <- matrixTricot[,mixedorder(colnames(matrixTricot))]
  matrixTricot <- as.matrix(matrixTricot)

  loglk <- apply(matrixTricot, 1, thurstonian_loglikehood)
  return(-1*sum(loglk,na.rm = T))
 }

 # Optimization parameters
 thurstone <- function(x){

  # Optimization parameters
  cp <- list(maxit = 10000, temp = 50, trace = TRUE, REPORT = 5000)

  # Initial values of worth values
  matMu <- runif(ncol(x),-5,5)
  names(matMu) <- paste0("Option", dimnames(x)[[2]])

  # Optimize to get the worth values
  # The first parameter is constraint to zero, therefore is discarded from the input
  opt <- optim(par = matMu[-1], tricotNLLMU0, x = x,
               method =c("Nelder-Mead","SANN","BFGS")[3], control = cp, hessian=TRUE)

```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
  # Extract worth values
  result <- c(0,opt$par)
  # Put the name of the first variable back
  names(result) <- dimnames(x)[[2]]

  #Return the result
  return(result)

}

# make simulation results reproducible
set.seed(123)

# Load and prepare data ------------------------------------------------------------
dfs <- map(1:11, readxl::read_excel,
           path = "Farmer_accuracy_2022.xlsx")

sheets <- c("farmer_demo",
            "E1_cocoa",
            "E1_maize",
            "E2_cocoa",
            "E2_maize",
            "E3_cocoa",
            "E3_maize",
            "E4_cocoa",
            "E4_maize",
            "E4_cocoa_proc",
            "E4_maize_proc")

dfs <- dfs %>% set_names(sheets)
list2env(dfs, globalenv())

# Convert some variables
farmer_demo$education <- match(farmer_demo$education, c("No formal Education", "Basic school", "Secondary school", "Junior high/ middle high school"))
farmer_demo$years_growing_cocoa <- match(farmer_demo$years_growing_cocoa, c("1-3", "3-5", "5-10", "10-20", ">20"))
farmer_demo$years_growing_maize <- match(farmer_demo$years_growing_maize, c("1-3", "3-5", "5-10", "10-20", ">20"))

```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
# Remove outlier
E3_maize <- E3_maize[-which(E3_maize$farmer_measurement > 40),]

# joining heaps cocoa and farmer demographic data
E2_cocoa <- E2_cocoa %>%
  inner_join(farmer_demo,
             by = c("location" = "location",
                    "farmer_id" = "farmer_id"))

# joining heaps maize and farmer demographic data
E2_maize <- E2_maize %>%
  inner_join(farmer_demo,
             by = c("location" = "location",
                    "farmer_id" = "farmer_id"))

# joining tin_can_cocoa and farmer demographic data
E3_cocoa <- E3_cocoa %>%
  inner_join(farmer_demo,
             by = c("location" = "location",
                    "farmer_id" = "farmer_id"))

# joining tin_can_maize and farmer demographic data
E3_maize <- E3_maize %>%
  inner_join(farmer_demo,
             by = c("location" = "location",
                    "farmer_id" = "farmer_id"))

# joining weighing_cocoa and farmer demographic data
E4_cocoa <- E4_cocoa %>%
  inner_join(farmer_demo,
             by = c("location" = "location",
                    "farmer_id" = "farmer_id"))


# joining weighing_maize and farmer demographic data
E4_maize <- E4_maize %>%
  inner_join(farmer_demo,
             by = c("location" = "location",
                    "farmer_id" = "farmer_id"))

```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
# Split exercise 4
E4_cocoa_untr <- subset(E4_cocoa, training=='Untrained')
E4_cocoa_tr <- subset(E4_cocoa, training=='Trained')

E4_maize_untr <- subset(E4_maize, training=='Untrained')
E4_maize_tr <- subset(E4_maize, training=='Trained')

# Set up Table 4+5
# Indexing will be by row and column name to avoid any mistakes
# Set up Table 4
Table4 <- matrix(NA, 10, 4)
varNames <- c(
  'JND',
  'RMSD_true',
  'MAD_true',
  'MSD_true')
colnames(Table4) <- varNames
rownames(Table4) <- c('E1M', 'E1C', 'E2M', 'E2C', 'E3M', 'E3C',  'E4MTr', 'E4CTr', 'E4MUntr',
'E4CUntr')

# Exercise 1 Maize
E1_maize_ranks <- matrix(NA,max(E1_maize$reps),max(E1_maize$ranks))
E1_maize_ranks[as.matrix(E1_maize[,c("reps", "ranks")])] <- E1_maize$farmer_ranked
worth_E1_maize <- thurstone(as.rankings(E1_maize_ranks))
dd_E1_maize                    <-                    as.data.frame(cbind(worth_E1_maize,
weight=sort(unique(E1_maize$actual_weight), decreasing=TRUE)))
lm_E1_maize <- lm(weight  ~ worth_E1_maize, data=dd_E1_maize)
0.6745 * sqrt(2*(coef(lm_E1_maize)[2]^2))

# Exercise 1 Cocoa
E1_cocoa_ranks <- matrix(NA,max(E1_cocoa$reps),max(E1_cocoa$ranks))
E1_cocoa_ranks[as.matrix(E1_cocoa[,c("reps", "ranks")])] <- E1_cocoa$farmer_ranked
worth_E1_cocoa <- thurstone(as.rankings(E1_cocoa_ranks))
dd_E1_cocoa                    <-                    as.data.frame(cbind(worth_E1_cocoa,
weight=sort(unique(E1_cocoa$actual_weight), decreasing=TRUE)))
lm_E1_cocoa <- lm(weight  ~ worth_E1_cocoa, data=dd_E1_cocoa)
0.6745 * sqrt(2*(coef(lm_E1_cocoa)[2]^2))

# JND as a way to compare exercises 1 and 2
```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
Table4["E1C", "JND"] <- calcJND(E1_cocoa, varnames=c("farmer_ranked", "ranks", "reps",
"actual_weight"), conv=FALSE)
Table4["E1M", "JND"] <- calcJND(E1_maize, varnames=c("farmer_ranked", "ranks", "reps",
"actual_weight"), conv=FALSE)

Table4["E2C", "JND"] <- calcJND(E2_cocoa, varnames=c("farmer_estimation", "ranks", "reps",
"actual_weight"), conv=TRUE)
Table4["E2M", "JND"] <- calcJND(E2_maize, varnames=c("farmer_estimation", "ranks", "reps",
"actual_weight"), conv=TRUE)
x <- E2_maize

# Exercises 2 to 4

# Deviance stats
E2statsC <- calcStats(E2_cocoa$farmer_estimation, E2_cocoa$actual_weight,
E2_cocoa$farmer_id)
Table4["E2C", names(E2statsC)] <- E2statsC
E2statsM <- calcStats(E2_maize$farmer_estimation, E2_maize$actual_weight,
E2_maize$farmer_id)
Table4["E2M", names(E2statsM)] <- E2statsM

E3statsC <- calcStats(E3_cocoa$farmer_measurement, E3_cocoa$actual_number_full,
E3_cocoa$farmer_id)
Table4["E3C", names(E3statsC)] <- E3statsC
E3statsM <- calcStats(E3_maize$farmer_measurement, E3_maize$actual_number_level,
E3_maize$farmer_id)
Table4["E3M", names(E3statsM)] <- E3statsM

E4statsC_untr <- calcStats(E4_cocoa_untr$farmer_estimation_kg,
E4_cocoa_untr$actual_weight_kg, E4_cocoa_untr$farmer_id)
Table4["E4CUntr", names(E4statsC_untr)] <- E4statsC_untr
E4statsM_untr <- calcStats(E4_maize_untr$farmer_estimation_kg,
E4_maize_untr$actual_weight_kg, E4_maize_untr$farmer_id)
Table4["E4MUntr", names(E4statsM_untr)] <- E4statsM_untr

E4statsC_tr <- calcStats(E4_cocoa_tr$farmer_estimation_kg, E4_cocoa_tr$actual_weight_kg,
E4_cocoa_tr$farmer_id)
Table4["E4CTr", names(E4statsC_tr)] <- E4statsC_tr
E4statsM_tr <- calcStats(E4_maize_tr$farmer_estimation_kg, E4_maize_tr$actual_weight_kg,
E4_maize_tr$farmer_id)
```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
Table4["E4MTr", names(E4statsM_tr)] <- E4statsM_tr

#Convert from cans to kg
Table4["E3C",] <- Table4["E3C",] * (64.3/50)
Table4["E3M",] <- Table4["E3M",] * (25/14)

#Convert to percentages
RefWeights <- c(25.0, 20.0, 25.0, 64.3, 25.0, 64.3, 2.5, 4.0, 2.5, 4.0)

#Values of Tables 4 and 5, absolute values and percentages of reference weights
round(Table4, digits=2)
round(sweep(Table4, 1, RefWeights, FUN = "/")*100, digits=1)


# Density plots ------------------------------------------------
create_density_plot <- function(input, var) {

  if (!(var %in% names(input))) {
    stop("Variable not found in the dataframe")
  }

  # Create the initial density plot
  p <- ggplot(input, aes(x = .data[[var]])) +
    geom_density() +
    xlab(NULL) +
    ylab(NULL) +
    theme_minimal() +
    theme(plot.title = element_text(size = 10, hjust = 0),
          axis.title.x = element_text(size = 8),
          axis.title.y = element_text(size = 8)) +
    scale_y_continuous(labels = scales::number_format(accuracy = 1), breaks = c(0, 12)) +
    scale_x_continuous(limits = c(-0.5, 0.5))  # Fix the x-axis limits

  # Add the normal distribution curve
  p <- p + stat_function(fun = dnorm,
                         args = list(mean = mean(input[[var]], na.rm = TRUE),
                                     sd = sd(input[[var]], na.rm = TRUE)),
                         color = "blue", size = 0.5)

  # Expand the limits (optional if scale_x_continuous is set)
```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
  p <- p + expand_limits(y = c(0, 12))

  return(p)
}

#Calculate relative error
E2_maize$relative_error <- (E2_maize$farmer_estimation - E2_maize$actual_weight) /
E2_maize$actual_weight
E2_cocoa$relative_error <- (E2_cocoa$farmer_estimation - E2_cocoa$actual_weight) /
E2_cocoa$actual_weight
E3_maize$relative_error <- (E3_maize$farmer_measurement - E3_maize$actual_number_level)
/ E3_maize$actual_number_level
E3_cocoa$relative_error <- (E3_cocoa$farmer_measurement - E3_cocoa$actual_number_full) /
E3_cocoa$actual_number_full
E4_maize_tr$relative_error <- (E4_maize_tr$farmer_estimation_kg -
E4_maize_tr$actual_weight_kg) / E4_maize_tr$actual_weight_kg
E4_cocoa_tr$relative_error <- (E4_cocoa_tr$farmer_estimation_kg -
E4_cocoa_tr$actual_weight_kg) / E4_cocoa_tr$actual_weight_kg
E4_maize_untr$relative_error <- (E4_maize_untr$farmer_estimation_kg -
E4_maize_untr$actual_weight_kg) / E4_maize_untr$actual_weight_kg
E4_cocoa_untr$relative_error <- (E4_cocoa_untr$farmer_estimation_kg -
E4_cocoa_untr$actual_weight_kg) / E4_cocoa_untr$actual_weight_kg

#Make plots
plot_list <- list(create_density_plot(E2_maize, "relative_error"),
                  create_density_plot(E2_cocoa, "relative_error"),
                  create_density_plot(E3_maize, "relative_error"),
                  create_density_plot(E3_cocoa, "relative_error"),
                  create_density_plot(E4_maize_tr, "relative_error"),
                  create_density_plot(E4_cocoa_tr, "relative_error"),
                  create_density_plot(E4_maize_untr, "relative_error"),
                  create_density_plot(E4_cocoa_untr, "relative_error"))

plot_names <- c("Ex2a. Maize - visual", "Ex2b. Cocoa - visual", "Ex3a. Maize - volumetric",
"Ex3b. Cocoa - volumetric", "Ex4a. Maize - scale, trained", "Ex4b. Cocoa - scale, trained", "Ex4c.
Maize - scale, untrained", "Ex4d. Cocoa - scale, untrained")

combined_plot <- (plot_list[[1]] + ggtitle(plot_names[1]) +
                    plot_list[[2]] + ggtitle(plot_names[2]) +
                    plot_list[[3]] + ggtitle(plot_names[3]) +
```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

```
                plot_list[[4]] + ggtitle(plot_names[4]) +
                plot_list[[5]] + ggtitle(plot_names[5]) +
                plot_list[[6]] + ggtitle(plot_names[6]) +
                plot_list[[7]] + ggtitle(plot_names[7]) +
                plot_list[[8]]) + ggtitle(plot_names[8]) +
                plot_layout(ncol = 2)
print(combined_plot)

```


Downloaded from https://cabidigitallibrary.org by 2a01:e0a:b94:2740:5cd:ea12:f4df:2ce2, on 10/30/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions